

VARSHITH CHENNURU

AI/ML Engineer • Agentic Systems, RAG & LLM Infrastructure

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PROFESSIONAL SUMMARY

AI/ML Engineer with a Master's in Computer Science and 3+ years building production LLM agent systems, RAG pipelines, and enterprise data platforms. Hands-on experience across the full agentic stack: multi-agent orchestration with LangChain/LangGraph, retrieval-augmented generation (embedding, chunking, hybrid search, reranking), and tool-use frameworks with guardrails, audit trails, and human-in-the-loop approvals. Background extends into language model development — has built small language models and is currently pretraining a large language model for Indic languages and Indian knowledge (~15-20 trillion token corpus), giving a working understanding of foundation model training alongside applied agent and RAG engineering. Fluent across Python, TypeScript/Go, cloud infrastructure (Azure/AWS/GCP), and CI/CD.

TECHNICAL SKILLS

Agentic AI & LLM Engineering: Multi-agent orchestration, LangChain, LangGraph, tool use / function calling, prompt engineering, context management, hallucination mitigation, guardrails & audit trails, Model Context Protocol (MCP)-style tool integrations

RAG & Retrieval: RAG pipeline design, embedding models, chunking strategies, hybrid vector + keyword search, reranking, FAISS, Qdrant, semantic code search (tree-sitter)

Language Model Development: Small language model (SLM) training and fine-tuning; currently pretraining a large-scale LLM for Indic languages/knowledge (~15-20T token corpus); foundation model architecture and evaluation

Programming: Python, TypeScript/JavaScript, Go, Java, SQL

Web & APIs: FastAPI, REST, GraphQL, React, Next.js, Microservices, WebSockets, n8n

Cloud & MLOps: Azure (ADF, Blob, Pipelines, DevOps), AWS, GCP, Docker, Kubernetes, CI/CD (Azure Pipelines, GitHub Actions), Terraform-style IaC

Data Engineering: Medallion Architecture (Bronze/Silver/Gold), ETL/ELT, Apache Airflow, CDC, Parquet

AI Coding Tools: Claude Code, GitHub Copilot, Cursor — used daily to accelerate development

PROFESSIONAL EXPERIENCE

Software Developer — AI/ML — EzSCM

Apr 2025 – Present

- Designed and deployed a generative AI onboarding agent using LangGraph, orchestrating multi-step, context-aware workflows and reducing manual onboarding effort by 70%.
- Built agent tool-use and dispatch logic connecting the LLM layer to backend services, with state tracking across multi-turn workflows.
- Integrated NLP-driven conversational AI into a React frontend, delivering adaptive, intent-aware interfaces that measurably improved user engagement.
- Containerized and deployed scalable ML microservices via Docker and Kubernetes, maintaining 99.9% uptime with automated failover.

Software Engineer — ML & Data — Krasan Consulting Services

Jun 2024 – Present

- Architected a Medallion (Bronze–Silver–Gold) data migration pipeline on Azure (ADF + Blob Storage + Azure SQL) ingesting 7 SACWIS tables (10M+ records) with row-count validation, Parquet partitioning, automated rollback, and full audit logging.
- Implemented Silver-layer cleaning (type casting, validity/dedup flags, SHA-256 SSN hashing, idempotent stored procedures) and Gold-layer MERGE-based upserts with surrogate keys and pipeline audit logs — patterns directly applicable to building crash-recovery and audit-trail behavior into agentic pipelines.
- Built an ML-powered ATS using Random Forest across 2,500+ profiles (~85% precision), cutting screening time ~30%; automated LLM/VLM document parsing of 100+ PDFs at 95% accuracy using Python + n8n.
- Created an AI-driven task generator (LangChain + Azure DevOps APIs) that reduced project planning time by 20%; designed CI/CD pipelines (Azure Pipelines, GitHub Actions) for automated ML deployments.

Software Engineer — ML — Hudson Insurance Group

Jun 2023 – Apr 2024

- Built a Python weather prediction model (ARIMA + Random Forest) on ~30 years of NOAA data, reducing insurance pricing errors by 15%; optimized SQL queries and React dashboards, improving data retrieval by 45%.

- Deployed real-time Azure event logging pipelines, reducing system downtime by 35% in high-stakes production environments.

Software Engineer — Krushna53, India

Apr 2022 – Jul 2022

- Implemented sentiment analysis and recommendation engines using GraphQL and Python APIs, boosting personalization and engagement by 25%.

KEY PROJECTS

Indic-LKM — Large Language Model for Indian Languages & Knowledge

Ongoing

- Pretraining a large-scale language model on a ~15-20 trillion token corpus spanning Indic languages and Indian domain knowledge, covering data curation, tokenizer design, and training pipeline design.
- Builds on prior hands-on experience training and fine-tuning smaller language models, applying that foundation-model background to inform grounding, evaluation, and failure-mode analysis in downstream agent and RAG systems.

DevRel Agent — 24/7 AI Developer Relations Platform

Dec 2025 – Apr 2026

- Built a multi-LLM developer support agent (Claude, GPT, Gemini) with FastAPI + Next.js supporting RAG, FAISS-based semantic code search (tree-sitter indexing), and incident debugging through a unified platform.
- Implemented guardrails with human-in-the-loop approvals, complete audit trails for compliance, and automated diagnosis for API latency issues and codebase knowledge discovery.

VoiceAge Platform — AI Customer Service Voice Agent

Mar 2026 – May 2026

- Built an intelligent phone automation platform combining Vapi voice infrastructure, OpenAI GPT-4o, and Qdrant vector DB for natural, real-time customer conversations with direct Zammad ticketing integration.
- Implemented real-time knowledge base retrieval via vector embeddings during live calls; post-call GPT-4o pipeline extracted sentiment/intent and auto-created enriched tickets with AI-generated tags.
- Designed a 7-collection Qdrant architecture (customer lookup, call history, analysis storage) with PII redaction, HMAC webhook auth, input sanitization, and SMS/email follow-up scheduling.

Agentic AI Browser Assistant — Chrome Extension

Sep 2025 – Nov 2025

- Developed a Chrome extension with autonomous agentic capabilities (deep analysis, web search, email composition, save/bookmark) using Gemini 3072-dimensional embeddings for semantic page indexing and RAG-based retrieval.
- Enabled cross-session browsing memory with intelligent section highlighting, context-aware search, adaptive theming, and export support (PDF, text, clipboard).

Medallion Architecture — Legacy Data Migration (Azure ADF)

2024 – 2025

- End-to-end Bronze–Silver–Gold pipeline processing 7 SACWIS tables (10M+ records); Silver layer with SHA-256 SSN hashing, dedup flags, and stored procedure cleaning; Gold MERGE upserts with surrogate keys and a 29-entry audit log.

Breast Cancer Detection (CNN) • ML-Powered ATS

2023 – 2024

- CNN trained on ~30,000 histology images achieving ~93% accuracy with 25% inference optimization. Random Forest ATS ranked 2,500+ candidate profiles at ~85% precision, deployed as a REST API.

EDUCATION

M.S. Computer Science (Engineering) — GPA 3.67

Sep 2022 – Jun 2024

Portland State University — Machine Learning, AI, Large Language Models, Data Engineering, Software Engineering, Algorithms

B.E. Computer Science — GPA 8.0/10

Jun 2018 – Jul 2022

Lendi Institute of Engineering and Technology

CERTIFICATIONS

Google Cloud Professional Data Engineer • Google Generative AI Leader